

Practical Work No. 11

Records

I. Definition of a Record

A record or structure is a way to group objects (variables), either of the same or different types, within a single entity identified by a single variable name. The variables contained in the structure are called fields of the structure. For a structure-type variable, the compiler allocates a memory area whose size is the sum of all the fields in the structure.

II. Declaration of a Record in C

Solution 1: Define a variable of record type.

```
struct Complex {  
    float real;  
    float img;  
};  
struct Complex C1;
```

Solution 2: Define a generic type for the record (a model) and then define the variable.

```
typedef struct {  
    float real;  
    float img;  
} Complex ;  
Complex C1;
```

Note: The type of each field can itself be a structure.

```
typedef struct {  
    int day;  int month;  int year;  
} Date;
```

```
typedef struct {  
    int code;  char lastName[20];  char firstName[20];  Date dob;  
} Student;
```

III. Using a Structure-Type Variable

Example: *Accessing fields of a structure*

```
struct Complex {  
    float real;  
    float img;  
};  
struct Complex c1;  
struct Complex *PC;
```

/ For a simple variable: fields are accessed using the dot operator (.) */*

```
c1.real = 5.2;
```

```
c1.img = 3.2;
```

/ For a pointer variable: fields are accessed using either the dot operator (.) or the special arrow operator (->) */*

```
PC = &c1;
```

```
(*PC).real = 4.4; /* Equivalent to PC->real = 4.4 */
```

IV. Memory Allocation for a Record

Example:

```
struct StudentRecord {
    char name[25];
    char firstName[25];
    int age;
    int studentID;
};
struct StudentRecord *p = NULL;
p = (struct StudentRecord *) malloc(sizeof (struct StudentRecord));
/* Allocates memory for: 25*sizeof(char) + 25*sizeof(char) + sizeof(int) + sizeof(int) */
...
free(p);
```

V. Application Exercises

Exercise 1

Write the C sub-programs to:

1. Input a complex number z1.
2. Display the real and imaginary parts of z1.
3. Display the real and imaginary parts of z2, which is the image of z1 under central symmetry.
4. Calculate the product of two complex numbers z1 and z2 passed as parameters.
5. Calculate the sum of two complex numbers z1 and z2 passed as parameters.

Use the following calculation formulas:

$$(a + b i) + (c + d i) = (a + c) + (b + d) i$$

$$(a + b i) * (c + d i) = (a * c - b * d) + (a * d + b * c) i$$

Exercise 2

Create an array TabEmp to store information about a group of employees in a company (ID, Name, Salary, Marital_Status). Fill it and display the number of employees whose salary is between 700 and 1500 DT.

Exercise 3

A phone directory consists of a set of pairs <name, phone> where name is the subscriber's name and phone is their number. Write a program to find the phone number of a given subscriber.

- **Solution 1:** Represent the directory as two parallel indexed arrays.
- **Solution 2:** Represent the directory using an array (or pointer) to a record composed of two fields.

Exercise 4

A circle is characterized by its center (of type Point) and its radius (of type float). Each circle can:

1. Provide the position of its center.
2. Provide the size of its radius.
3. Move.
4. Resize (the radius must remain strictly positive).
5. Calculate its surface area.
6. Calculate its perimeter.
7. Provide all its characteristics (including the surface area and perimeter).

Define the data structures and sub-programs to:

1. Create a circle C, display its characteristics, move it to the origin, set its radius to 1, and then display its characteristics again.
2. Create a point P in an arbitrary location and create two concentric circles C1 and C2 with different radii, both having P as their center. Display the position of circle C1, move C2 with given steps, and then display the position of C1 again.